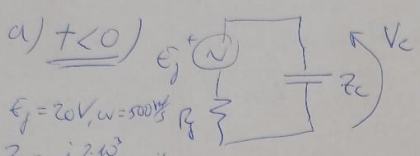
PEI 1 (Recuperación) 15-06-2021

P1)

a) $t<0$



$\mathcal{E}_j = 20V, \omega = 500 \frac{rad}{s}$

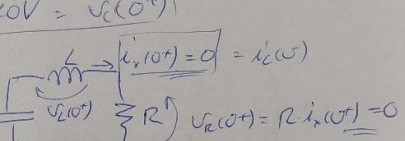
$Z = -j\frac{2\omega^3}{500} = -j4\Omega$

$V_c = \frac{\mathcal{E}_j}{R_g + Z_c} \cdot Z_c = (40 - 10j)V$

$\Rightarrow v_c(t) = 10\sqrt{2}\cos(500t - \frac{\pi}{4})V$

$\Rightarrow \boxed{v_c(t=0^+) = 10V = v_c(0^+)}$

en $t = 0^+$



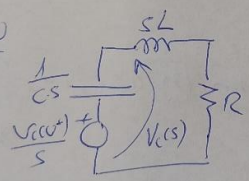
$\Rightarrow \boxed{i_x(0^+) = 0} = i_c(0)$

$v_c(0^+) = 10V$

$v_R(0^+) = R \cdot i_x(0^+) = 0$

$\Rightarrow \boxed{v_L(0^+) = v_c(0^+) - R i_x(0^+) = 10V}$

b) $t>0$



$\boxed{V_L(s) = \frac{\frac{v_c(0^+)}{s}}{R + Ls + \frac{1}{Cs}}(R + Ls)}$

$= \boxed{\frac{v_c(0^+)(s + \frac{R}{L})}{s^2 + \frac{R}{L}s + \frac{1}{LC}}}$

c) Polin. caract: $s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$

$\Rightarrow s^2 + 1000s + 10^6 = 0$

$\Rightarrow s_{1,2} = \frac{-1000 \pm \sqrt{1000^2 - 4 \cdot 10^6}}{2} = -500 \pm j\frac{1}{2}\sqrt{3} \cdot 500$

$\Rightarrow$ Ingen amortiguado: $v_c(t) \sim e^{-500t}\sin(\sqrt{3}500t + \phi)$

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$\omega_0 = \sqrt{\frac{1}{LC}} = \sqrt{10^6} = 1000 \frac{rad}{s}$

$2\xi\omega_0 = \frac{R}{L} \Rightarrow R = 2L\xi$

para $\xi = 1 \Rightarrow \boxed{R = 4\Omega}$ Amortiguado crítico

d)

$V_c(s) = \frac{10(s+10^3)}{s^2 + 10^3 s + 10^6} = \frac{10(s+10^3)}{(s+500)^2 + 75 \cdot 10^4}$

$s_{1,2} = 500(-1 \pm \sqrt{3}j)$ $\quad$ $Re[s_1]$ $\quad$ $(Im[s_1])^2$

$V_c(s) = \frac{1}{75} \cdot \frac{10^4 \cdot 75}{(s+500)^2 + 75 \cdot 10^4} + \frac{10 \cdot s}{(s+500)^2 + 75 \cdot 10^4}$

$\omega_0^2 = 75 \cdot 10^4$, $\alpha = 500$

$a = 0$, $\alpha = 500$, $\omega_0^2 = 75 \cdot 10^4$

para $t>0$

$v_c(t) = \frac{1}{75}e^{-500t}\sin(\sqrt{75 \cdot 10^4}t) + 10\frac{10^3}{\sqrt{75 \cdot 10^4}}e^{-500t} \cdot \sin(\sqrt{75 \cdot 10^4}t - 1{,}047)$ V.

para $t<0$

$v_c(t) = 10\sqrt{2}\cos(500t - \frac{\pi}{4})V.$

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P2)

a) $\begin{cases} V_1 = h_{11} I_1 + h_{12} V_2 \\ I_2 = h_{21} I_1 + h_{22} V_2 \end{cases}$



medido con $V_2 = 0$: $h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0} = Z_{in} = (1+j)\,\Omega$

$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0} = \frac{-I_{cc}}{I_{g1}} = -\frac{2}{3}$

medido con $V_1 = 0$ $\begin{cases} 0 = h_{11}(-I_{cc}) + h_{12} E_{g2} \\ I_2 = h_{21}(-I_{cc}) + h_{22} E_{g2} \end{cases}$

$\Rightarrow h_{12} = \frac{h_{11} I_{cc}}{E_{g2}} = -\frac{3}{2}$

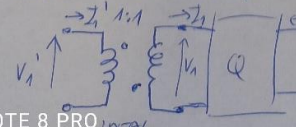
$h_{22} = \frac{\frac{E_{g2}}{Z_g} + I_{cc} h_{21}}{E_{g2}} = 1\,\Omega^{-1}$

$\Rightarrow h(Q) = \begin{pmatrix} (1+j)\,\Omega & -\frac{3}{2} \\ -\frac{2}{3} & 1\,\Omega^{-1} \end{pmatrix}$

b) Para determinar $i_x(t)$ asociamos los cuadripolos a través de los parámetros "h" al estar conectados serie - paralelo y al ser las corrientes de entrada cero debido al transformador ideal.

Q'

Param. "h" del cuadripolo de "arriba" cambian:



$\begin{cases} V_1 = h_{11} I_1 + h_{12} V_2 \\ I_2 = h_{21} I_1 + h_{22} V_2 \\ V_1' = -V_1 \\ I_1' = -I_1 \end{cases}$ y !!

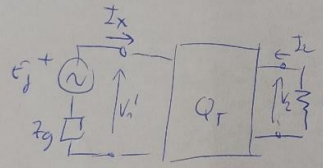
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$\begin{cases} -V_1' = h_{11}(-I_1') + h_{12} V_2 \\ I_2 = h_{21}(-I_1') + h_{22} V_2 \end{cases} \Rightarrow \begin{cases} V_1' = h_{11} I_1' - h_{12} V_2 \\ I_2 = -h_{21} I_1' + h_{22} V_2 \end{cases}$

$h(Q') = \begin{pmatrix} h_{11} & -h_{12} \\ -h_{21} & -h_{22} \end{pmatrix}$

$\Rightarrow$ Asociam : $h(Q_T) = h(Q) + h(Q')$

$h(Q_T) = \begin{pmatrix} 2h_{11} & 0 \\ 0 & 2h_{22} \end{pmatrix}$



$\begin{cases} V_1' = E_g - Z_g I_x \\ V_1' = 2h_{11} I_x + 0 \\ I_2 = 0 + h_{22} V_2 \\ V_2 = -R_L I_2 \end{cases}$ $\}$ (x°)

$\Rightarrow I_x = \frac{V_1'}{2h_{11}} = \frac{E_g - Z_g I_x}{2h_{11}} \Rightarrow I_x = \frac{E_g}{Z_g + 2h_{11}} = 1\,A$

$\Rightarrow \boxed{i_x(t) = \cos(\omega t)\,A}$

c) $\boxed{P_{R_L} = 0}$ pues para que la corriente no sea $I_2 = 0 = V_2$.

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